mand SHALL be rejected with CONSTRAINT_ERROR if the Power, Duration, or Cause values are not permitted as above, or with FAILURE otherwise.

Start of adjustment

The ESA SHALL generate a PowerAdjustStart Event when it begins to adjust its power.

The ESA then begins to adjust its power consumption or generation to the power level commanded in the Power field.

End of adjustment (normal completion)

After the elapsed duration, the ESA SHALL revert to normal (or idle) power levels.

The ESA SHALL also generate a PowerAdjustEnd Event with a cause code to indicate a 'Normal completion', the ESAState SHALL be restored to Online, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

Failure or Opt-out during adjustment

If during the power adjustment session a failure or other condition occurs (such as the user deciding to opt-out by updating the OptOutState) then the ESA SHALL generate a PowerAdjustEnd Event to indicate the end of the session, with the appropriate cause code.

The ESAState SHALL be updated to reflect the new state, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

Receiving a new PowerAdjustRequest command when adjustment is on-going

If an existing power adjustment is already happening, and a new PowerAdjustRequest command is received, then if the ESA allows it, the ESA SHALL return SUCCESS, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value from the Cause field of the new command. If the ESA does not permit this new PowerAdjustmentRequest command to interrupt the adjustment that is in progress, it SHALL return BUSY.

Note that if the new command is accepted, then the ESA SHALL NOT generate a new PowerAdjustEnd Event until the new duration has elapsed. This is to avoid generating too many events in cases where a device is routinely commanded with overlapping power adjustments.

For example, a battery inverter ESA may be sent a new request every 5 seconds to adjust its discharge power based on real-time meter readings. Each command may have a 60 second duration, but this command is superseded after 5 seconds by a new request.

After the client has sent its last command and after this command duration expires then the last active power adjustment has been completed. This final command causes the PowerAdjustEnd Event to be generated when the ESAState is restored to Online, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

9.2.9.2. CancelPowerAdjustRequest Command

Allows a client to cancel an ongoing PowerAdjustmentRequest operation.

